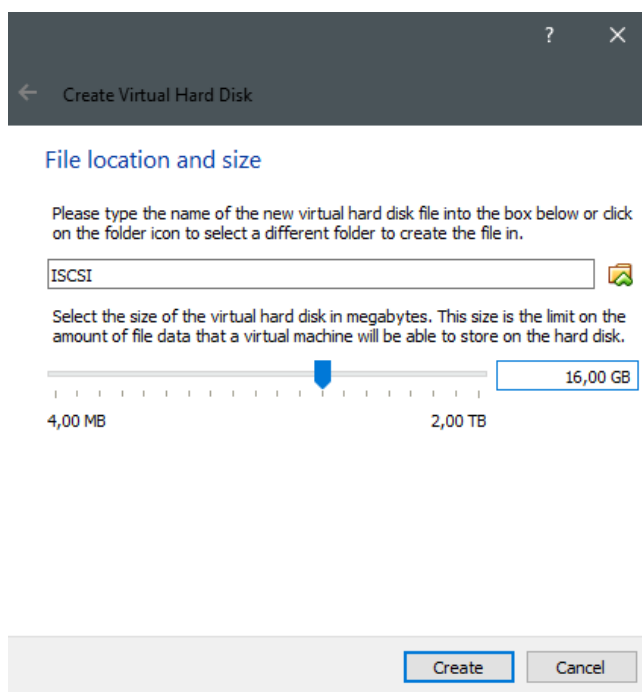


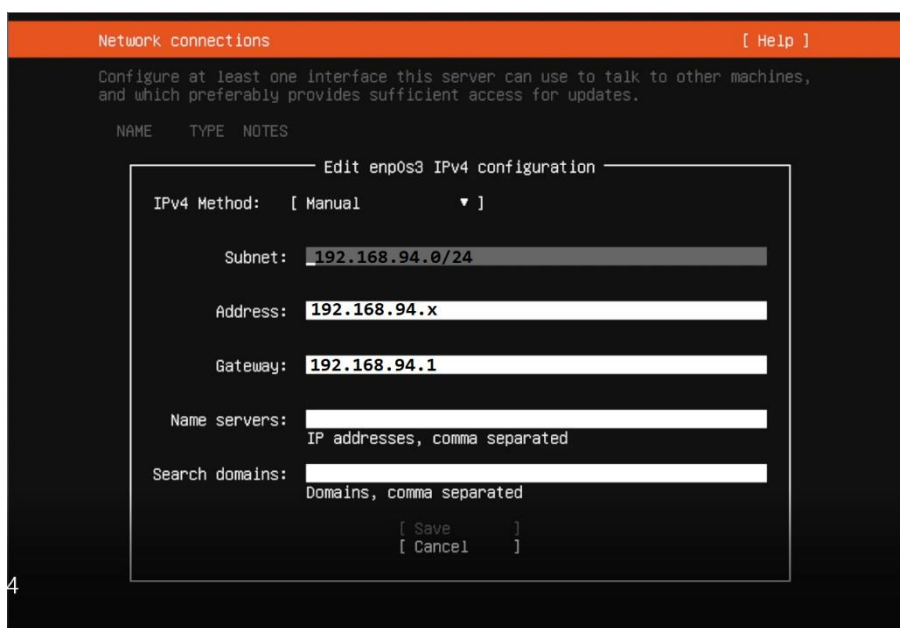
Laborator ISCSI

Ubuntu server 18.04

Începem prin a crea o mașină virtuală cu plăcile de rețea în modul **Bridged** și cu un Hard Disk de **minimum 16 GB**. Sistemul de operare este Ubuntu 18.04 server și va trebui instalat în mașina virtuală.



Atentie, la realizarea configurării internetului se vor introduce următoarele:



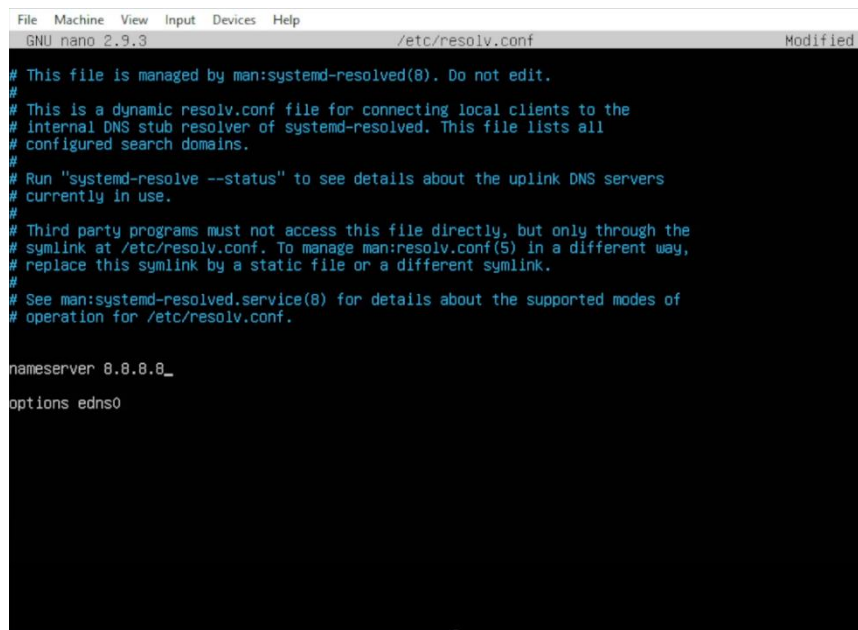
Unde x va fi ales in cadrul laboratorului astfel incat sa nu dea conflict cu celelalte adrese IP din laborator.

După instalare se va introduce următoarea comandă in **modul administrator**:

sudo apt-get update

In fișierul **resolv.conf** se afla informații despre serverele DNS necesare.

nano /etc/resolv.conf



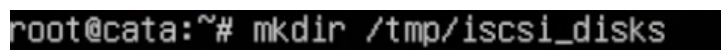
```
File Machine View Input Devices Help
GNU nano 2.9.3 /etc/resolv.conf Modified

# This file is managed by man:systemd-resolved(8). Do not edit.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "systemd-resolve --status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs must not access this file directly, but only through the
# symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a different way,
# replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 8.8.8.8_
options edns0
```

apt -y install targetcli-fb

mkdir /tmp/iscsi_disks



```
root@cata:~# mkdir /tmp/iscsi_disks
```

Se porneste aplicatia pentru configurare ISCSI targetcli.

Atentie comenzile date in aceasta aplicatie sunt precedate de **/>** iar cele din line de comanda a sistemului de operare de **#!** In aplicatia targetcli functioneaza comenzile uzuale de **cd** si **ls** pentru a naviga in structura de directoare.

targetcli

```
root@cata:~# targetcli
Warning: Could not load preferences file /root/.targetcli/prefs.bin.
targetcli shell version 2.1.fb43
Copyright 2011-2013 by Datera, Inc and others.
For help on commands, type 'help'.

/> ls
0- / ..... [...]
  0- backstores ..... [Storage Objects: 0]
    0- block ..... [Storage Objects: 0]
    0- fileio ..... [Storage Objects: 0]
    0- pscsi ..... [Storage Objects: 0]
    0- ramdisk ..... [Storage Objects: 0]
  0- iscsi ..... [Targets: 0]
  0- loopback ..... [Targets: 0]
  0- vhost ..... [Targets: 0]
/> _
```

Exista mai multe tipuri de stocare pentru ISCSI:

An administrator can choose any of the following backstore devices that Linux-IO (LIO) supports:

fileio backstore

Create a `fileio` storage object if you are using regular files on the local file system as disk images. For creating a `fileio` backstore, see [Creating a fileio storage object](#).

block backstore

Create a `block` storage object if you are using any local block device and logical device. For creating a `block` backstore, see [Creating a block storage object](#).

pscsi backstore

Create a `pscsi` storage object if your storage object supports direct pass-through of SCSI commands. For creating a `pscsi` backstore, see [Creating a pscsi storage object](#).

ramdisk backstore

Create a `ramdisk` storage object if you want to create a temporary RAM backed device. For creating a `ramdisk` backstore, see [Creating a Memory Copy RAM disk storage object](#).

In acest laborator vom folosi tipul fileio, adica un fisier pe HDD:

backstores/fileio create iscsi_file /tmp/iscsi_disks/disk01.img 1G

```
/> backstores/fileio create iscsi_file /tmp/iscsi_disks/disk01.img 1G
Created fileio iscsi_file with size 1073741824
```

iscsi/

```
/> iscsi/
```

create iqn.2018-08.com.example.storagesrv01.targe01

```
/iscsi> create iqn.2018-08.com.mere.storagesrv01.target01
Created target iqn.2018-08.com.mere.storagesrv01.target01.
Created TPG 1.
Global pref auto_add_default_portal=true
Created default portal listening on all IPs (0.0.0.0), port 3260.
```

iqn.2018-08.com.example.storagesrv01.targe01/tpg1/

```
/iscsi> iqn.2018-08.com.mere.storagesrv01.target01/tpg1/
```

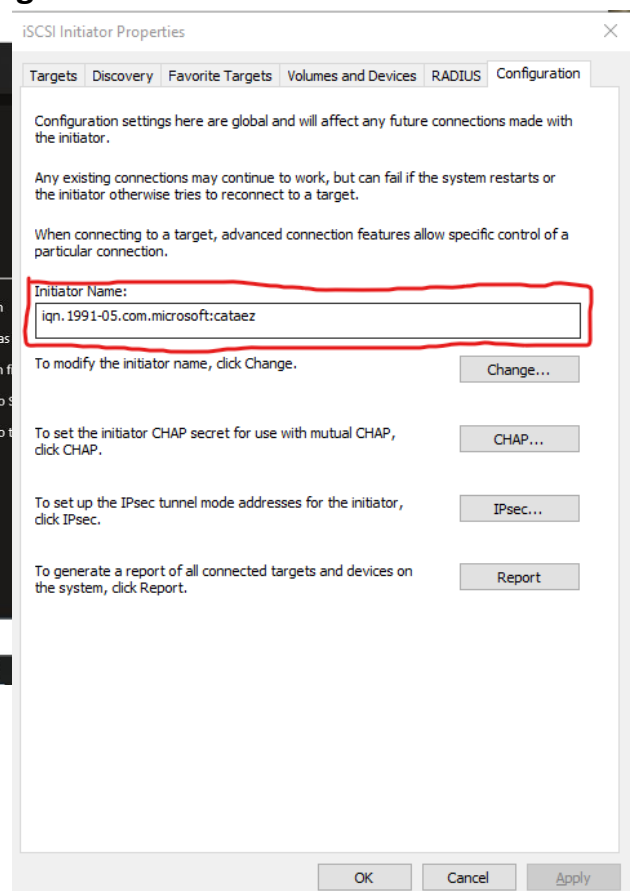
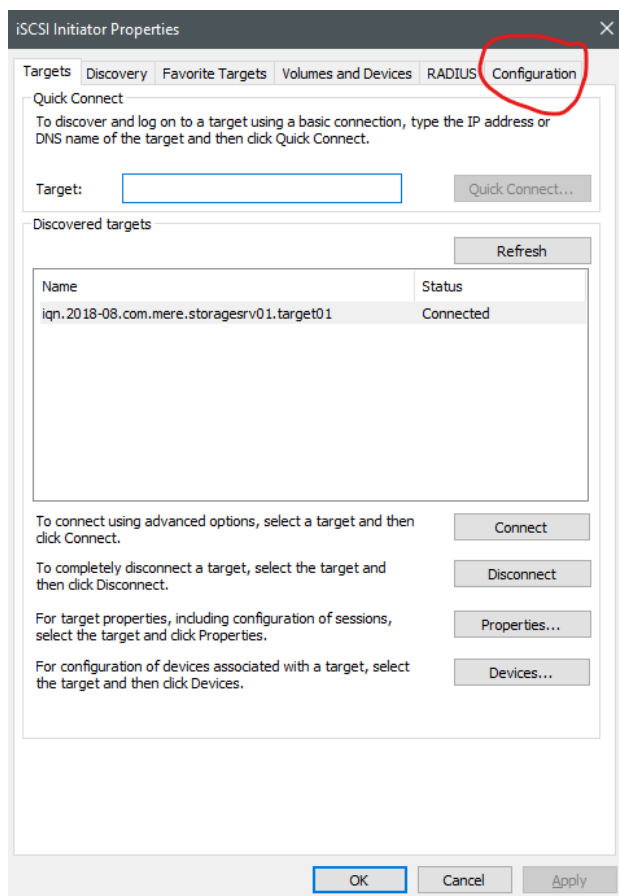
luns/ create /backstores/fileio/iscsi_file

```
/iscsi/iqn.20...target01/tpg1> luns/ create /backstores/fileio/iscsi_file
Created LUN 0.
```

acls/

```
/iscsi/iqn.20...target01/tpg1> acls/
```

Create „initiator name”
Acest initator name se gaseste astfel

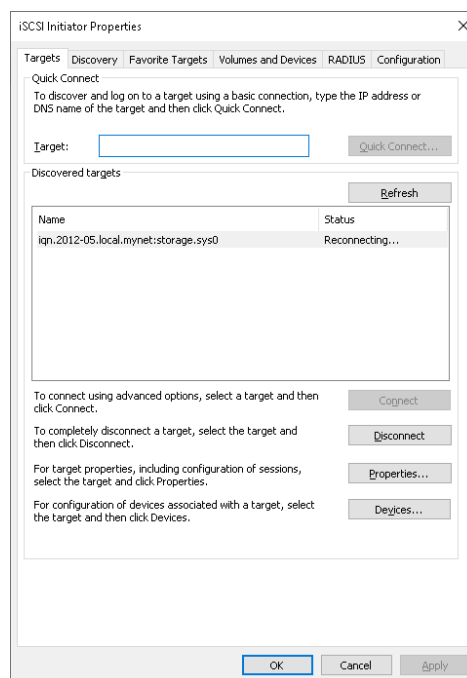


```
/iscsi/iqn.20...t01/tpg1/acls> create iqn.1991-05.com.microsoft:cataez
Created Node ACL for iqn.1991-05.com.microsoft:cataez
Created mapped LUN 0.
```

exit

```
/iscsi/iqn.20...t01/tpg1/acls> exit
Global pref auto_save_on_exit=true
Last 10 configs saved in /etc/rtplib-fb-target/backup.
Configuration saved to /etc/rtplib-fb-target/saveconfig.json
```

La final se realizează conectarea cu ajutorului iSCSI Initiator-ului din Windows la IP-ul masinii virtuale anterior creata.

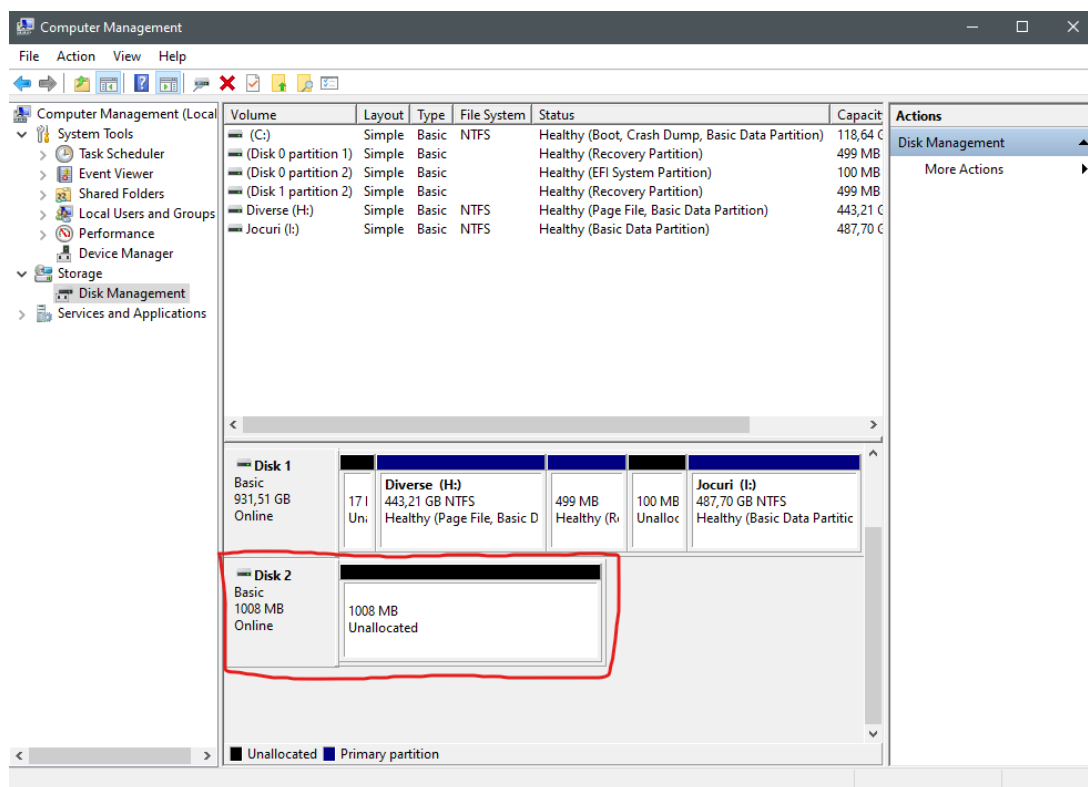


Pentru a nu interveni probleme, este recomandata dezactivarea Firewall-ului cand incercati conectarea sau permiterea portului 3260 .

Se apasa butonul *Connect* si in cazul in care conectarea a avut succes apare la status **Connected** ,

In continuare, pentru a configura o partitie pe noul disc conectat prin iSCSI, trebuie accesata fereastra Computer Management (Click dreapta This PC > Manage > Storage)

Aici se poate observa o partite de exact 1GB nealocata.



Anexa:

Alte detalii despre procesul de instalare le găsiți aici:

https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9-beta/html/managing_storage_devices/configuring-an-iscsi-target_managing-storage-devices